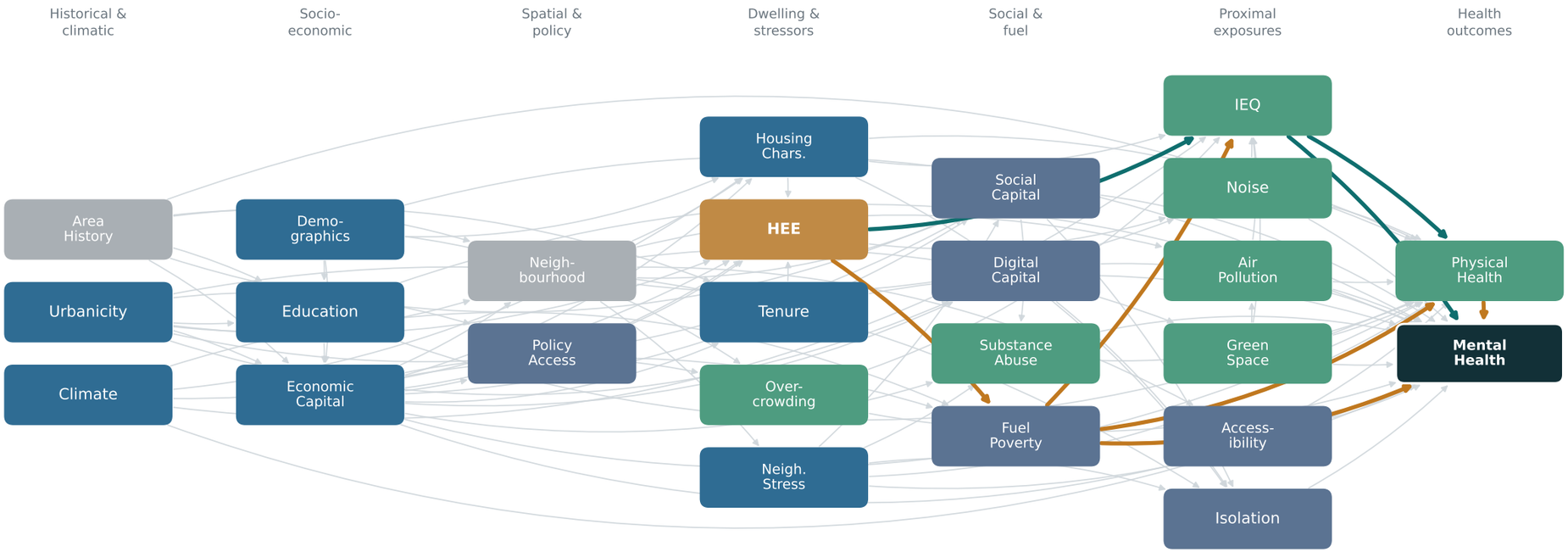
A. The reviewed DAG — 25 constructs, 89 directed edges



Legend

- Exposure (HEE)
- Outcome (MH)
- Structural confounder
- Environmental mediator
- Social / structural mediator
- Latent (unobserved)

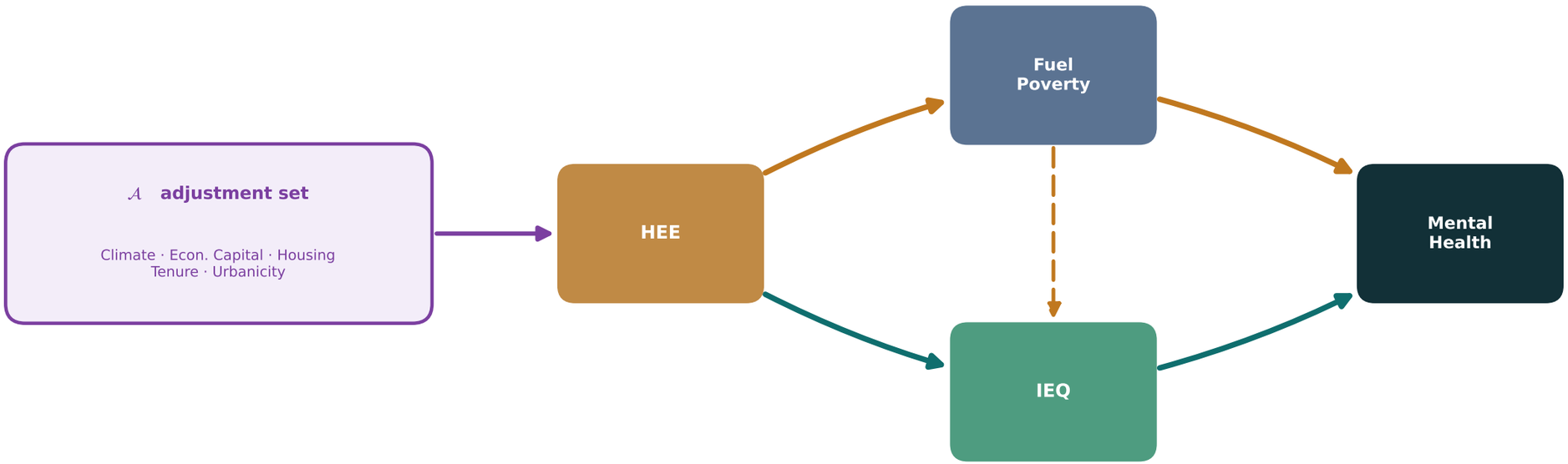
Highlighted pathways

1 HEE → Fuel Poverty → MH
affordability of adequate warmth

2 HEE → IEQ → MH
warmth, damp and mould

Minimal adjustment set $\mathcal{A}$
Climate · Economic Capital · Housing characteristics · Tenure · Urbanicity, returned by the backdoor criterion in dagitty.

B. The two pathways, isolated



Pathway 1 — affordability

efficient fabric lowers the cost of adequate warmth

measured at LSOA level (LILEE) — testable

Pathway 2 — indoor environment

efficient fabric changes warmth, damp and mould

no area-level measure — untestable here

No HEE → MH edge: the DAG asserts the effect is fully mediated, and $\mathcal{A}$ blocks every backdoor path. The dashed arrow is the crossover added at expert review — heating rationing under fuel poverty also degrades the indoor environment.